

Internal Standard Method to Measure Nickel Nitrate Concentration with Fluorescence

Spectroscopy

Name: Duy Hoang

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Purpose

The goal of this experiment was to determine the concentration of an unknown Ni^{2+} solution using an internal standard calibration method. Co^{2+} served as the internal standard and was added to all samples at a constant concentration. By plotting the ratio of Ni/Co absorbance versus the ratio of Ni/Co concentration for prepared standards, a calibration curve was generated and used to determine the Ni^{2+} concentration in an unknown sample.

Results and Discussion

The goal of this experiment was to determine the true concentration of an unknown Ni^{2+} solution using an internal standard (IS) method. Cobalt(II) (Co^{2+}) was added at constant concentration to all standards and the unknown, and the ratio of Ni^{2+} absorbance to Co^{2+} absorbance was used to correct for instrumental variation. A calibration plot of signal ratio (Ni/Co) versus concentration ratio (Ni/Co) was generated, and the concentration of the unknown Ni^{2+} solution was calculated from the regression line.

The calibration curve yielded a linear fit with $R^2 = 0.9934$, indicating strong correlation between concentration ratio and signal ratio. Blank-corrected absorbance values ensured accuracy in calculating response ratios.

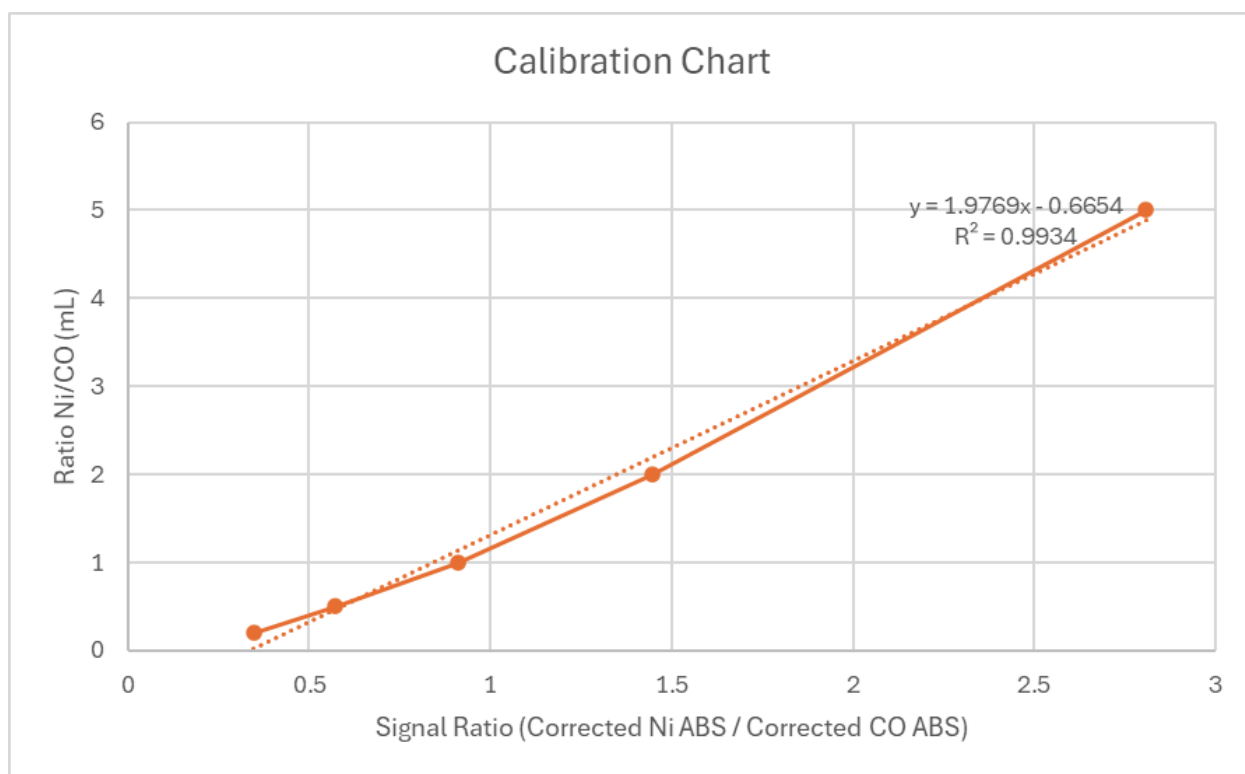
From replicate measurements of the unknown solution, the mean Ni^{2+} concentration was 0.1630 M, with a standard deviation of 0.02803 M. The RSD was 2.93%, indicating acceptable precision ($\text{RSD} \leq 3\%$ threshold). The true concentration provided by the instructor was 0.200 M.

Therefore, the absolute error was: $|0.1630 - 0.2000| = 0.03706$ M and the relative error was:
 $0.03706 / 0.2000 \times 100 = 18.52\%$

Since the relative error exceeds 10%, the measurement accuracy is not acceptable. This suggests systematic error rather than random error, supported by the relatively low RSD.

Likely contributors to reduced accuracy include pipetting bias, incomplete mixing of standards or unknown, cuvette contamination, or insufficient re-blanking of the instrument. Accuracy may be improved by using calibrated Class-A glassware, verifying pipette volumes, thoroughly mixing solutions, wiping cuvettes before each run, and re-zeroing the spectrophotometer between measurements.

Overall, while the calibration curve demonstrated good linearity and the precision of the measurement was acceptable, the large systematic error indicates that method execution must be improved to obtain an accurate result.



Concentration of Ni(NO ₂) ₂	Standard deviation of concentration (M)	RSD of concentration (%)	Absolute error (M)	Relative error (%)
0.162951646	0.028028897	2.930579187	0.037048354	18.52417707

Questions

- 1.) The internal standard calibration curve displayed strong linearity with an R^2 value of 0.9934, indicating a strong correlation between the concentration ratio and signal ratio. Although not perfectly linear, the high R^2 suggests the internal standard method functioned correctly and the data reasonably fit a first-order calibration model.

- 2.) The measured mean Ni^{2+} concentration of the unknown was 0.1630 M, compared to the true value of 0.200 M. This resulted in an absolute error of 0.03706 M and a relative error of 18.52%. Because this relative error exceeds 10%, the accuracy of the measurement is considered not acceptable.

Potential contributors to poor accuracy include pipetting variability when preparing standards and the unknown, incomplete mixing, instrument drift, and cuvette variation (fingerprints or bubbles). Accuracy may be improved by using Class A volumetric glassware, recalibrating pipettes, consistently wiping cuvettes, mixing thoroughly, and re-blanking more frequently.

- 3.) The standard deviation for replicate measurements was 0.02803 M, and the RSD was 2.93%, which meets the $\leq 3\%$ criterion for acceptable precision. This result indicates that while the measurements were not accurate, they were precise, suggesting systematic error rather than random error. Improving technique and calibration quality would likely correct the bias without negatively impacting precision.